Notes on NAND topology MPROM delay

Luca Seggiani
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We want to study the delay experienced in a NAND topology **MPROM** (*Mask-Programmable Read Only Memoery*). Of course, in reality this will be due to wire capacities and resistances, in addition to capacities and resistances tied to MOS transistors.

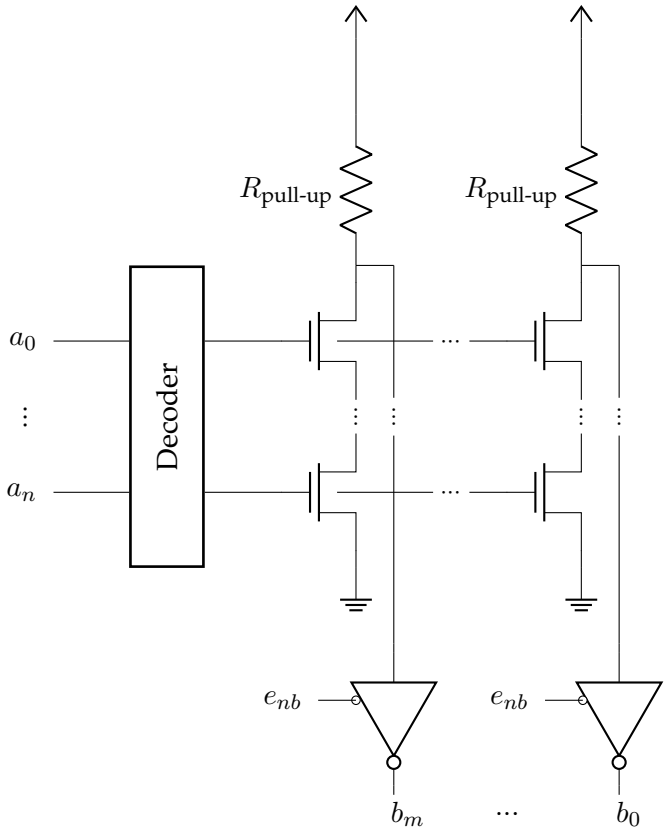
1 Simplified model

In this discussion we consider the simplified model:

- No wire capacity or resistance will be considered;
- A simplified model of an NMOS transistor is used, in which only the drain-source channel resistance (R_d) and capacitance (C_d).

2 NAND topology

Let us first consider a NAND topology memory, which we will base further considerations on.



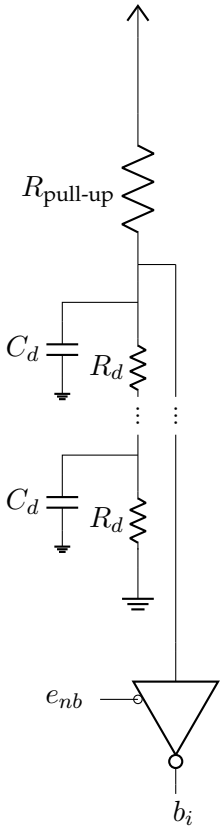
The horizontal lines at the NMOS gates are the **word lines** WL_0 ... WL_n. The vertical lines to the inverters are the **bit lines** BL_0 ... BL_m.

The **decoder**, in the NAND topology, keeps every line but the selected one at logic *high*. A "1" at a certain position in the memory can thus be encoded by shorting the channel of the corresponding MOS, ensuring conduction to ground happens even when that MOS is disabled (that is, selected).

3 Elmore delay

We model the delay in the network through **Elmore delay**. The idea is that the total delay in an RC network is given by the sum of delays measured at each capacitor.

For our discussion, we focus on the delay of a single bit line.



We don't consider any MOS to be selected, and instead model the intrinsic channel resistance R_d and channel capacitance C_d of each one.

We start by considering the delay measured at each capacitor i, with $i = 0$ being the top capacity, and i_n being the bottom one. Since each capacitor sees the resistance above it, we get the following relation on time characteristics:

$$\begin{aligned}\tau_0 &= R_{\text{pull-up}} C_d \\ \tau_1 &= (R_{\text{pull-up}} + R_d) C_d \\ \tau_2 &= (R_{\text{pull-up}} + 2R_d) C_d\end{aligned}$$

Or more concisely:

$$\tau_i = (R_{\text{pull-up}} + iR_d) C_d$$

From this we get the total delay as the sum of all capacitor delays along the bit line:

$$\tau_{\text{tot}} = \sum_{i=0}^{n} \tau_i = \sum_{i=0}^{n} (R_{\text{pull-up}} + iR_d) C_d$$

At this point we consider the effect from the pull-up resistor $R_{\text{pull-up}}$ and the intrinsic capacitor resistors R_d separately:

$$\tau_{\text{tot}} = R_{\text{pull-up}} \cdot nC_d + \sum_{i=0}^{n} iR_dC_d$$

At this point it's useful to define the following *equivalent* capacitances and resistances:

$$C_{eq} = nC_d$$
$$R_{eq} = nR_d$$

These, after substituting in the equation above, give:

$$\tau_{\text{tot}} = R_{\text{pull-up}} \cdot C_{eq} + \sum_{i=0}^{n} i\frac{R_{eq}C_{eq}}{n^2} = R_{\text{pull-up}} \cdot C_{eq} + \frac{R_{eq}C_{eq}}{n^2}\sum_{i=0}^{n} i$$

At this point all that's left is applying (one of many) Euler's formula:

$$\tau_{\text{tot}} = R_{\text{pull-up}} \cdot C_{eq} + \frac{R_{eq}C_{eq}}{n^2}\frac{n(n-1)}{2} = R_{\text{pull-up}} \cdot C_{eq} + \frac{R_{eq}C_{eq}}{n}\frac{(n-1)}{2}$$

This result becomes more useful in the limit of infinity, so:

$$\lim_{n\to+\infty} \tau_{\text{tot}} = \lim_{n\to+\infty} R_{\text{pull-up}} \cdot C_{eq} + \frac{R_{eq}C_{eq}}{n}\frac{(n-1)}{2} = R_{\text{pull-up}} \cdot C_{eq} + \frac{1}{2}R_{eq}C_{eq}$$

Which at the left hand side is the standard result from Elmore delay.